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ORIGINAL ARTICLE

Vision-based initial point alignment control for the wheel hubs in the robotic polishing system

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Abstract

In the robotic polishing system for the wheel hubs, initial point alignment is a very important procedure before the robot starts to polish. Since the surface shapes of the wheel hubs are complex, traditional contact sensors are not suitable for the initial point alignment. Therefore, this paper presents a vision-based initial point alignment control method due to the merits of non-contact and high precision of the vision sensor. The main procedures of the new proposed initial point alignment methods are as follows. Firstly, the center point of the wheel hub is computed based on the growth of sampled points (GSP) method. Secondly, the center lines of the spokes of the wheel hub are extracted based on the edges of the spokes. Based on the center point and the center lines of the spokes, the needed rotation angle of the wheel hub denoting the current and target initial points can be computed. Thirdly, an alignment controller is designed to align the current initial point with the target one. Finally, experiments are well conducted to demonstrate the effectiveness of the proposed initial point alignment control method for the wheel hubs.

Keywords Vision · Robotic polishing system · Alignment control · Wheel hub · Initial point

1 Introduction

Wheel hubs are the important parts of the motor vehicles, which have critical impacts on the safety, comfort, and appearance of the motor vehicles [1]. Generally, the manufacturing of the wheel hub consists of five main processes, i.e., smelting, casting with low pressure, heat treatment, polishing, electroplating, or coating [2, 3]. Among the five processes, polishing is the most important factor affecting the final quality and appearance of the wheel hub. However, at present, polishing is almost accomplished manually. The polishing efficiency is low, and the consistency of polishing quality cannot be guaranteed. Moreover, the poor polishing environment is harmful to the workers. To solve the problem, robotic polishing system for the wheel hubs needs to be designed.

Compared with the manual polishing, the robotic polishing system has several obvious advantages such as high polishing efficiency, high polishing precision, and good flexibility. Several robotic polishing systems have been designed. Tian [4] designed a robotic polishing system for curved surfaces, in which an industrial robot with six degrees of freedom (DOFs) and a six-dimensional force/torque sensor was used. Wang [5] developed a self-determination polishing robot finishing large mold free-form surface. A robot with four uniform distributed wheels was used which had two driving wheels and two driven wheels. The new small mobile robot was more suited for large mold surface. Ahmed [6] used an industrial robot with six DOFs for the freeform surface polishing. An automatic planning and programming system based on CAD data was proposed. In Ref. [7], a polishing system containing a macro-mini robot was presented, in which the mini-robot was used for the force control while the macro-robot was for the positioning the mini-robot.

In designing the robotic polishing systems, robot polishing path planning and force control are mainly studied [8–15]. Lin [9] proposed a polishing trajectory generator based on cutter data generated from the postprocessor of a CAD system. Liu [10] presented a path planning method in which a complex part is partitioned into simple surfaces

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such as relatively flat surface and curved surface, and a path connection method was designed. In Ref. [8], a new type of six-axis force/torque sensor was presented with overload protection and decoupling of force structurally. Zhang [11] designed a constant force actuator for industrial robot polishing and proposed a force feedback control method based on force sensor to keep the polishing contact force in a relatively constant range by dynamically adjusting the output force of the cylinder. To improve the polishing effect of the robot, some sensors have also been introduced into the robotic polishing systems. In Ref. [16], surface roughness evaluation was made during the robot-assisted polishing process based on analysis of the acoustic emission signals in the frequency domain. In Ref. [17], the sensor was employed for online assessment of workpiece surface roughness during robot-assisted polishing of steel bars.

In the abovementioned robotic polishing systems, sensors are only used for polishing quality assessment. In fact, automatic alignment of the initial points of the workpieces based on some kinds of sensors is also required, which will affect the subsequent robot polishing path directly and the final polishing quality. Although there are few literature reports on initial point alignment in the robotic polishing system, some researchers have proposed vision-based initial point alignment methods in the welding robot. Fang [18] proposed an initial point alignment method based on the geometric relationship between the reference and current seam trajectories. Zhu [19] proposed a template matching method to align the welding torch to the initial welding point. Since the surface shapes of the wheel hubs are complex and are quite different from the geometry of the welding seams, the initial point alignment methods mentioned above cannot be used in the robotic polishing system.

In this paper, an initial point alignment method using a vision sensor is proposed for the wheel hubs in the robotic polishing system. There are two reasons for choosing vision sensor. Firstly, the shape of the wheel hub is generally complex. It is not convenient and difficult to measure the initial point by other methods. Secondly, the vision sensor has the merits of non-contact, large amount of information, and high precision. The main procedures of the proposed initial point alignment control method are as follows. The difference between the current and target initial points of the wheel hub is denoted as the needed rotation angle of the wheel hub, which can be computed by the center point and the center lines of the spokes of the wheel hub. Therefore, the method to precisely detect the center point and the center lines of the spokes is firstly presented. Then, a closed-loop initial point alignment control system is designed to align the current initial point to the target one.

The rest of this paper is organized as follows. Section 2 describes the configuration of the robotic polishing system and the three stages of the robotic polishing process. The detailed image feature detection and angle computation of the wheel hub are given in Sections 3 and 4. In Section 5, the alignment controller is designed. Section 6 illustrates the experimental results and Section 7 concludes this paper.

2 The configuration of the robotic polishing system

The robotic polishing system for wheel hubs is shown in Fig. 1, which consists of a vision sensor, two polishing robots, two tool cabinets, a main rotator, and three sub-rotators. The vision sensor is composed of a camera and a light source. It measures the initial points of the wheel hubs, computes initial point alignment adjustment, and sends that to the alignment control system. The two robots are used for polishing the wheel hubs. One robot is used for rough polishing while the other for fine polishing. In the polishing process, multiple tools are required to be used alternately. Therefore, each robot is assigned a tool cabinet. The robot can replace the tools automatically under pneumatic driving. The main rotator is used for the alternation of the three workstations, i.e., vision measurement workstation (workstation A), rough polishing workstation (workstation B), and fine polishing workstation (workstation C). Each workstation has a sub-rotator. At the workstation A, the sub-rotator is used for initial point alignment based on the vision measurement, which rotates the wheel hubs from the current initial points to the target ones. At the workstation B and workstation C, the sub-rotators are used to rotate the wheel hubs at high speed to match the robot's polishing operation.

In order to better understand the robotic polishing system, the main work stages of the robotic polishing system for the wheel hubs are described. As shown in Fig. 2, there are three main work stages of the robotic polishing process for the wheel hubs, i.e., initial point alignment, rough polishing, and fine polishing. The alternation of the three work stages is realized through the main rotator. The unpolished wheel hubs are placed on the floor or in the workpiece basket at will. When they are put on the sub-rotator at the workstation A, the initial point is inconsistent with the target initial point, which will affect the follow-up robot polishing operations. Therefore, a vision-based initial point alignment method is proposed in this paper. It measures the differences between the current initial points and the target ones, computes the needed alignment adjustment, and sends the adjustment signals to the control system. After the initial points of the wheel hubs are precisely aligned, the subsequent rough and fine polishing actions are started. Since off-line robot programming method is adopted in this robotic polishing system, the robot polishing trajectory is kept the same as long as the type of the wheel hubs remains unchanged.

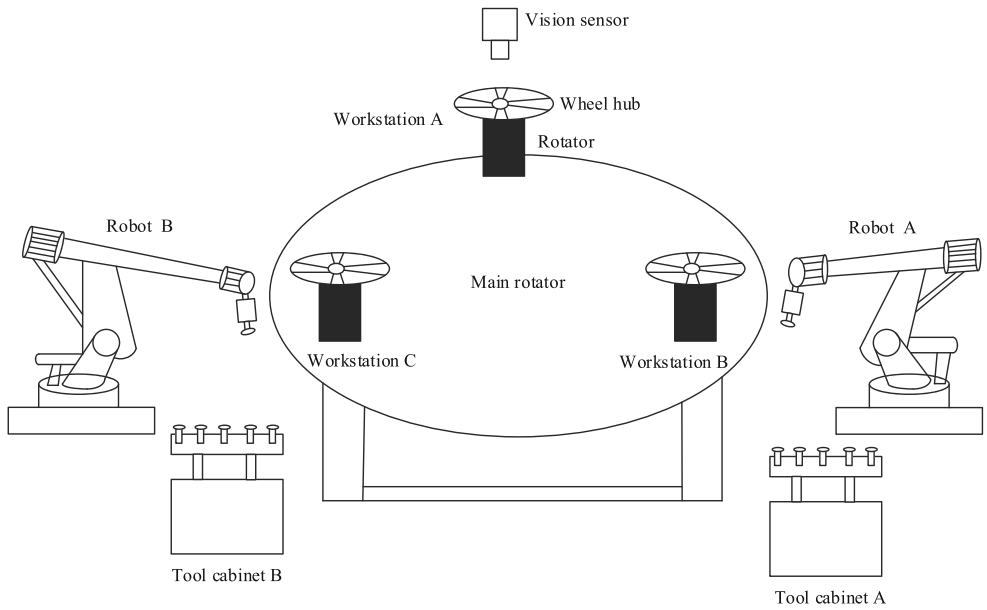


Fig. 1 The configuration of the robotic polishing systems for the wheel hubs

3 Image feature detection for the wheel hub

The difference between the current and target initial points can be denoted as the rotation angle between the two initial points. To compute the needed rotation angle of the wheel hub, a vision sensor is adopted. The vision sensor is composed of a camera and a light source. Since the wheel hub has certain reflectivity, a light source with parallel light is used. The rotation angle of the wheel hub is determined by its center and spokes. Therefore, the center of the wheel hub is firstly detected. Then, the center lines of the spokes are detected. The detailed extraction of the image feature detection for the wheel hub is given at follows.

3.1 Image pre-processing

Since the wheel hub is made of aluminum alloy, there are reflections on the surface of the wheel hub under the light source. To improve the quality of the image, median filter is firstly applied to the image, which can remove the salt and pepper noise while keeping edges unobscured. Then, image morphology erode method is used to delete the large disturbance such as light spot or white lines. Finally, a Gaussian filter is adopted to smooth the gray value to improve the quality of the image further. For the specific explanation and operation process of the median filter, erode operation, and Gaussian filter, refer to Ref. [20].

3.2 Detection of the center of the wheel hub

The center of the wheel hub needs to be detected precisely since it has crucial effect on the precision of the rotation angle of the wheel hub. To better detect the center of the wheel hub, the binary image is firstly obtained from the raw image using OTSU threshold algorithm as follows:

$$I_{f_b} = \begin{cases} 255, & \text{if } I_{f_r} \geq I_t \\ 0, & \text{others} \end{cases} \tag{1}$$

where I_{f_b} and I_{f_r} are the gray values of the binary image f_b and the raw image f_r, respectively, and I_t is the threshold gray value of the image computed using the OTSU algorithm [21].

Since the wheel hub is put on the spin axis of the sub-rotator and fixed firmly through the pneumatic head, the center of the wheel hub can be roughly determined

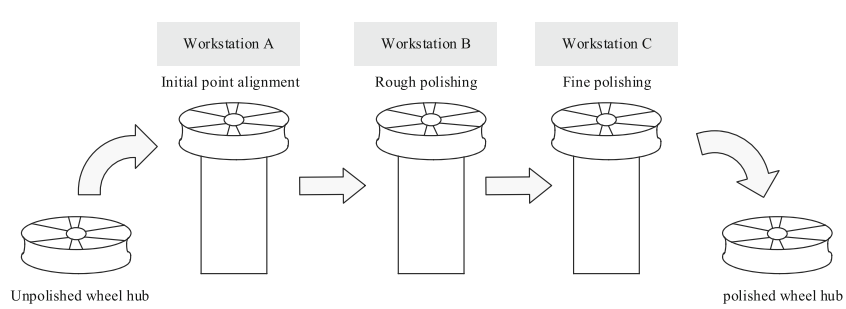


Fig. 2 The three main work stages of the robotic polishing system for the wheel hubs

beforehand, which is denoted as (x_c, y_c). To compute the precise center of the wheel hub, a method named growth of sampled points (GSP) is proposed. The main advantages of the new GSP method to detect the center of the wheel hub are high efficiency and good stability. The reasons are as follows. Firstly, the center point can be computed using only one single seed point and searching in four directions with GSP method while traditional methods like RANSAC uses all of the points and searches in every small angles. Secondly, the averaging operation of the center points computed from several seed points plays the role of smoothing filtering, which can improve the stability of the final computed center point. The description of the GSP is as follows.

In the circle area around the rough center (x_c, y_c), an image point is randomly sampled as:

$$(x^*, y^*) = \underset{\delta_c}{rand}(x, y) \quad (x, y) \in \delta_c \tag{2}$$

where (x^*, y^*) are the image coordinates of the sampled point, $rand()$ is the random function, and δ_c is the circle area defined by the rough center (x_c, y_c).

Based the sampled points, the GSP algorithm can be defined, which denotes that the new points grow at the horizontal and vertical directions in the current image from the sample points as follows:

$$\Delta L_i = \begin{cases} 1, & \text{if } I_{f_b}(x^* + i - 1, y^*) = 255 \text{ and } I_{f_b}(x^* + i, y^*) = 0 \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

$$\Delta R_i = \begin{cases} 1, & \text{if } I_{f_b}(x^* + i + 1, y^*) = 255 \text{ and } I_{f_b}(x^* + i, y^*) = 0 \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

$$\Delta U_i = \begin{cases} 1, & \text{if } I_{f_b}(x^*, y^* + i - 1) = 255 \text{ and } I_{f_b}(x^*, y^* + i) = 0 \\ 0, & \text{otherwise} \end{cases} \tag{5}$$

$$\Delta D_i = \begin{cases} 1, & \text{if } I_{f_b}(x^*, y^* + i + 1) = 255 \text{ and } I_{f_b}(x^*, y^* + i) = 0 \\ 0, & \text{otherwise} \end{cases} \tag{6}$$

$$L_s = \sum_{i=0}^{i=N_L} \Delta L_i \quad (x^* + N_L, y^*) \in \delta_c \tag{7}$$

$$R_s = \sum_{i=0}^{i=N_R} \Delta R_i \quad (x^* + N_R, y^*) \in \delta_c \tag{8}$$

$$U_s = \sum_{i=0}^{i=N_U} \Delta U_i \quad (x^*, y^* + N_U) \in \delta_c \tag{9}$$

$$D_s = \sum_{i=0}^{i=N_D} \Delta D_i \quad (x^*, y^* + N_D) \in \delta_c \tag{10}$$

where ΔL_i, ΔR_i, ΔU_i, and ΔD_i are the flags denoting the gray value of the image point i is 255 or 0. L_s, R_s, U_s, and D_s are the summations of the corresponding flags at the left, right, upward, and downward directions, respectively. N_L, N_R, N_U, and N_D are grown length at the left, right, upward, and downward directions, respectively.

To guarantee the reliability of the above GSP algorithm, the following verification is needed. If the following conditions are satisfied, the grown area is really the center area of the wheel hub.

$$\begin{cases} 0 < L_s + R_s \le D_{wt},\ 0 < U_s + D_s \le D_{wt} \\ L_s > 0,\ R_s > 0,\ U_s > 0,\ D_s > 0 \end{cases} \tag{11}$$

where D_{wt} is the diameter threshold value of the center area of the wheel hub.

It is well known that the perpendicular bisector of the chord of a circle passes through the center of the circle. Therefore, the coordinates of the center of the wheel hub denoted as (x_o, y_o) can be given by:

$$\begin{cases} x_o = x^* + (R_s - L_s)/2 \\ y_o = y^* + (D_s - U_s)/2 \end{cases} \tag{12}$$

Due to the noise influence of the random selected seed point (x^*, y^*) and computation bias of the L_s, R_s, U_s, and D_s, the accuracy of the computed center point cannot be guaranteed. To improve the accuracy of the center of the wheel hub, N sample seed points are selected and the corresponding centers are computed using Eqs. 3 to 12. Then, the final improved center is computed as the mean of the centers based on each sample seed point, which can be given by:

$$\begin{cases} \bar{x_o} = \frac{1}{N} \sum_{i=1}^{N} x_{oi} \\ \bar{y_o} = \frac{1}{N} \sum_{i=1}^{N} y_{oi} \end{cases} \tag{13}$$

where $(\bar{x_o}, \bar{y_o})$ is the final improved center of the wheel hub, and (x_{oi}, y_{oi}) is the center of the wheel hub computed based on the i-th sample seed point. In this paper, the specific value of N can be chosen based on that the fluctuation of the calculated center point is limited to a certain range. From our experiments, the value of N can be chosen as 15–30 (Fig. 3).

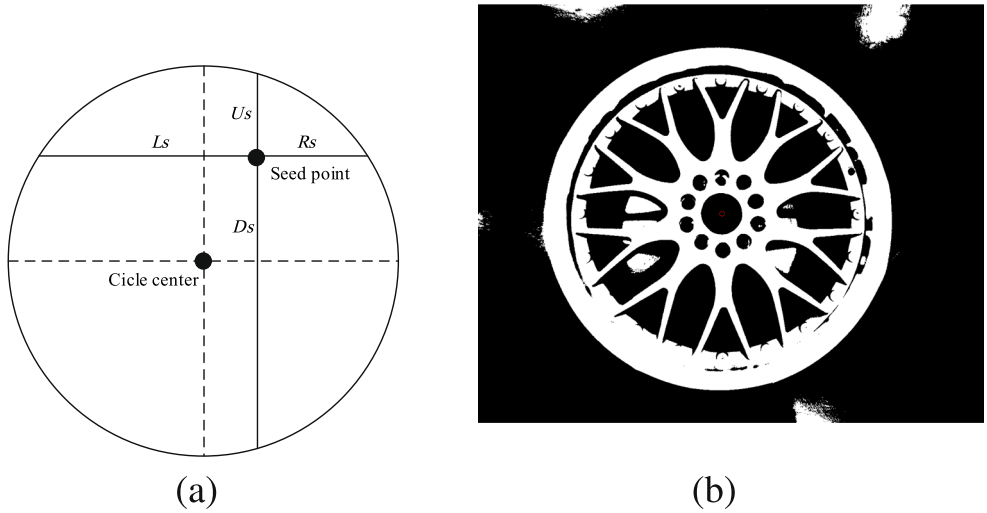


Fig. 3 The illustration of the GPS and detected circle center of the wheel hub: **a** GPS, **b** detected center of the wheel hub

3.3 Detection of the center lines of the spokes

The rotation angle of the wheel hub is determined by the center and the spokes. Therefore, after the center of the wheel hub is computed, the center lines of the spokes need to be extracted. The center lines of the spokes are determined by their left and right edges. The light reflection of the rotators and wheel hub will form bright spots in the captured images, which will have a bad influence on the edge detection of the spokes. Thus, the bright spots should be eliminated from the image firstly. In this paper, the bright spots are determined and eliminated based on the bright connections and image morphologic method. Firstly, the bright spots with smaller area than that of the spokes of the wheel hub are deleted as follows:

$$S_v = \{S_k \mid A_{Sk} > A_{St} \; and \; I(i, j) = 255, (i, j) \in S_k\} \quad (14)$$

where S_v is the set of the valid bright connections whose bright areas are bigger than a threshold value, S_k is the k-th bright connection, A_{Sk} is the area of the bright connection S_k, and A_{St} is the threshold value of the area of the bright connection.

The bright spots with a small area that are not connected with the spokes of the wheel hub can be deleted using Eq. 14. However, the bright spots connected with the spokes of the wheel hub cannot be deleted using the same method. These bright spots are deleted using the image morphologic method as given by:

$$E(I) = \{z \mid (B)_z \in I\} \quad (15)$$

where $E(I)$ is the erode operation to the image I, B is the structure element, which is designed according to the shape and the size of the bright spot, and z is the translation size of the structure element in the image.

After the bright spots are deleted from the image, the center lines of the spokes can be detected. In this paper, the center lines of the spokes are computed from the left and right edges of the spokes, which are based on the gray value distribution around the spokes. The gray values on the spokes are more bigger than those off the spokes, which is shown in Fig. 4. Therefore, the left and right edge points of the spokes are detected by:

$$E_l = \arg\min_k (I_{k-2} + I_{k-1} - I_k - I_{k+1}) \quad (16)$$

$$E_r = \arg\max_k (I_k + I_{k+1} - I_{k-1} - I_{k-2}) \quad (17)$$

where E_l and E_r are the left and right edge points of the spoke, respectively, I_i is the gray value of the i-th image point, and $i(i = k - 2 \cdots k + 2)$ is the i-th image points in the circle S_c, S_c is the set of the image points on the circle around the spoke. Then, the center point C_l of the spoke can be computed based on the left and right edge points as follows:

$$C_l = \frac{E_l + E_r}{2} \quad (18)$$

The center line of the spoke can be defined as the connection line between the center point of the wheel hub and the center point of the spoke. The extracted center lines of all the spokes are shown in Fig. 5. It can be seen that the center lines of the spokes are well extracted with the proposed method.

4 Computation of the angle of the wheel hub

From one spoke of the wheel hub, the angle of the wheel hub can be computed. However, due to the light reflection and rotation interference, the accuracy cannot be guaranteed. In this paper, the angle of the wheel hub is computed from all

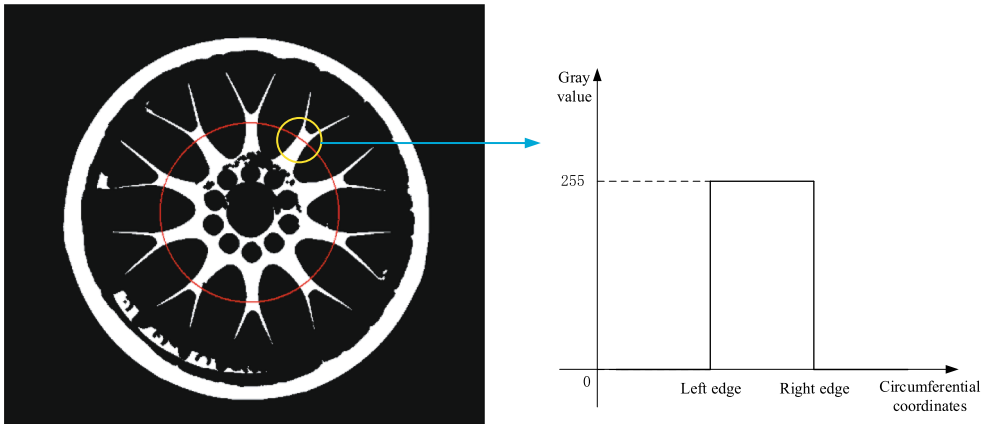


Fig. 4 The gray value distribution around the spoke

of the center lines of the spokes. Suppose the i-th center line of the spoke is described by:

$$a_i x + b_i y = 1, \quad (i = 1, 2, \cdots, n_{sp}) \tag{19}$$

where a_i and b_i are the parameters of the i-th center line of the spoke, and n_{sp} is the number of the spokes.

The angle θ_i of the i-th spoke is computed as:

$$\theta_i = \begin{cases} \arctan(-a_i/b_i), & \text{if } b_i \neq 0 \\ \pi/2, & \text{others} \end{cases} \tag{20}$$

The mean value of the angle calculated from each spoke of the wheel hub is considered as the final angle value of the wheel hub as follows:

$$\theta_{wh} = \frac{1}{n_s} \sum_{i=1}^{n_{sp}} \theta_i \tag{21}$$

where θ_{wh} is the angle of the wheel hub, and n_{sp} is the number of the spokes.

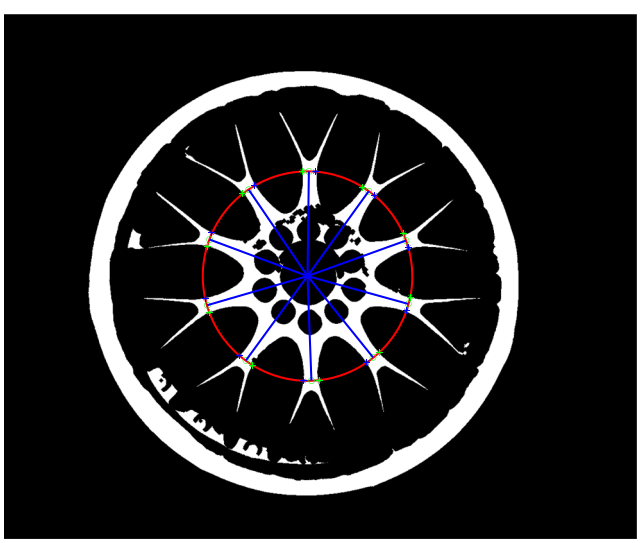


Fig. 5 The extracted center lines of the spokes

5 Design of the initial point alignment controller

The diagram of the closed-loop initial point alignment controller for the wheel hub in the robotic polishing system is shown in Fig. 6, which consists of two closed control loops, i.e., the vision control loop and the motor control loop. The vision control loop is used to detect the initial point of the wheel hub and align it with the target one. The purpose of the motor control loop is to make the running angle of the motor follow the target one. The motor control loop is realized in the motor servo driver. The vision control loop consists of two channels such as feedback and forward channels. The feedback channel computes the current initial point of the wheel hub based on the vision sensor. It consists of the feature extraction and the periodicity processing parts. The forward channel computes the adjustment of the sub-rotator holding the wheel hub based on the errors between the current and target initial point of the wheel hub. It consists of the controller and output verification parts. Since the feature extraction has been described above, the periodicity processing, controller design, and output verification parts will be given in the following sections.

5.1 The periodicity processing

Generally, the wheel hub has several spokes. The angle between each two spokes is the same. According to the periodical distribution of the spokes, the computed angle of the wheel hub based on the vision sensor can be processed before it is used in the controller. Suppose the number of the spokes is n_s, the adjusting angle between the current and target initial point must be smaller than θ_p, where $\theta_p = 2\pi/n_s$. As shown in Fig. 7, θ_r is the angle between the target initial point and the positive x-axis. The target initial point is set beforehand according to the optimized polishing path of the robot. θ_{f1} is the computed angle between the current initial point and positive x-axis with the vision sensor. θ_f is

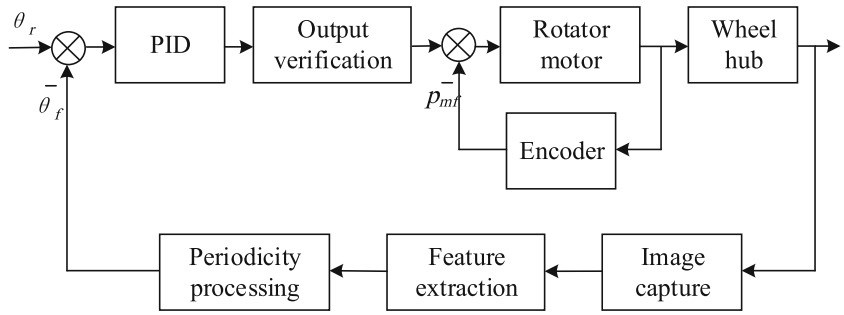


Fig. 6 The initial point alignment controller for the wheel hub in the robotic polishing system

the processed angle based on the periodicity of the spokes. The angle error sent to the controller can be processed as follows:

$$\theta_a = \begin{cases} \theta_e & \text{if } 0 \le \theta_e \le \theta_p \\ \theta_e - \lfloor \theta_e/\theta_p \rfloor \theta_p & \text{if } \theta_p < \theta_e \le 2\pi \end{cases} \tag{22}$$

where θ_e and θ_a are the angles between the current and target initial points of the wheel hub before and after the periodicity processing, respectively. θ_e is the error between θ_r and θ_{f1}, which can be computed as:

$$\theta_e = \theta_{f1} - \theta_r \tag{23}$$

5.2 Controller design

To guarantee the precision of the initial point alignment, closed-loop control framework is adopted. The control model named as "look-and-move" is adopted in the aligning process. The computed current initial point of the wheel hub is compared with the target one. If the difference between them is larger than a threshold, the alignment control continues. On the other hand, if the difference is smaller than a threshold, the alignment control for the current wheel hub stops and waits for the next wheel hub. The initial point alignment control is finished within some control sample times. The current output of the controller is only related to the state of the previous control sample time, but not to the initial state. Therefore, the incremental proportional-integral-differential (PID) controller is used in the initial point alignment control, which is expressed as:

$$\begin{aligned} u(k) = \lambda\{&K_p[e(k) - e(k-1)] + K_i e(k) \\ &+ K_d[e(k) - 2e(k-1) + e(k-2)]\} \end{aligned} \tag{24}$$

where $u(k)$ is the output of the controller at the k-th sample time, λ is the mapping factor from the angle value in the image space to the amount of the rotation of the motor, which can be well calibrated off-line, $e(k)$ is the error between the current and target initial points of the wheel hub, and K_p, K_i, and K_d are the proportional, integral and differential coefficients, which can be initially tuned using critical proportional band (CPB) method in the robot off-line programming system and fine adjusted on-line manually according to the performance of the initial point alignment control.

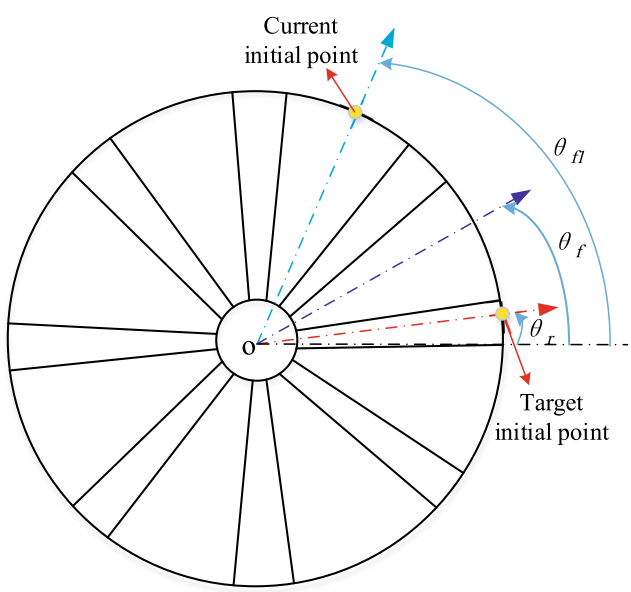


Fig. 7 The periodicity processing for the computed angle of the wheel hub

5.3 Output verification

The visual measurement may contain noise data due to the reflect light of the wheel hub or vibration of the rotator during the alignment process. To guarantee the smoothness and safety of the alignment control, the output of the controller needs to be verified before it is sent to the motor of the rotator. Two factors are considered in the output verification. On the one hand, since the difference between the current and target initial points of the wheel hub is not too large, the total output of the controller in the alignment control process needs to be confined to a specific range. On the other hand, the output of the controller at each

Fig. 8 The configuration of the wheel hub polishing system

sample time is also limited to a threshold value. The output verification of the controller can be described as follows:

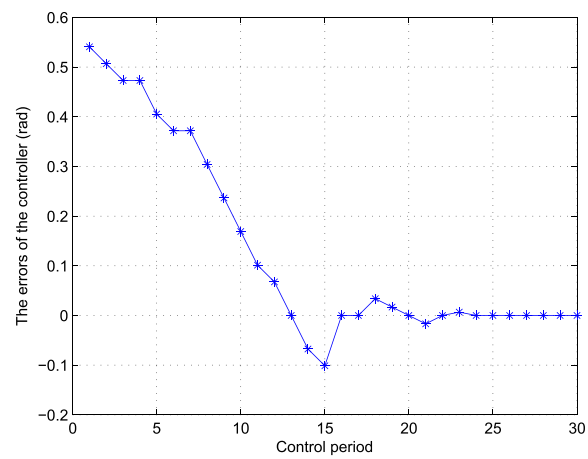
$$u_o(k) = \begin{cases} u(k), & \text{if } |u(k)| < u_t \text{ and } |T_u| < T_t \\ u_t, & \text{if } u(k) \geq u_t \text{ and } |T_u| < T_t \\ -u_t, & \text{if } u(k) \leq -u_t \text{ and } |T_u| < T_t \\ 0, & \text{if } |T_u| > T_t \end{cases} \tag{25}$$

where $u_o(k)$ is the verified output of the controller, u_t and T_t are the threshold values of the controller at each sample time and in the whole alignment process, respectively.

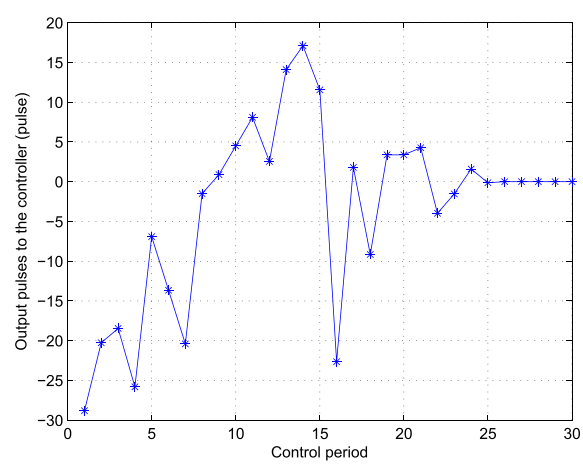
6 Experiments and results

To test the effectiveness of the proposed vision-based initial point alignment control method for the wheel hubs, experiments were well carried out. The experimental setup is shown in Fig. 8. The two polishing robots were ABB IRB4400-60. The main rotator was Weiss CR0700C. The rotator A, rotator B, and rotator C were self-manufactured, which were driven by servo-motors. The vision sensor was composed of a light source and a digital industrial camera. The light source was FG-THZ338318, which can emitted directional light. The camera was DH-HV3151UC-M, which captured images through the USB port. The image processing was implemented in industrial computer TPC7000, whose main frequency was 3.5 GHz, and RAM was 4G. The controller of the initial point alignment was implemented in Omron PLC CP1H. The vision measurement of the wheel hub was sent to the PLC from the industrial computer through RS232 port.

The parameters used in the feature extraction of the wheel hub were as follows: the size of the image was 1280 × 1024 in pixels, $D_{wt} = 900$, $N = 20$, $n_{sp} = 10$,



(a)



(b)

Fig. 9 The errors and output of the controller in the initial point alignment control process: **a** the errors of the controller, **b** the output of the controller

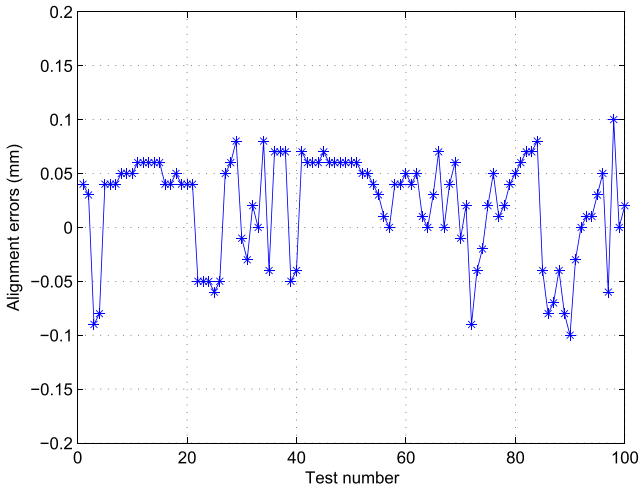


Fig. 10 The results of the precision evaluation of the proposed initial point alignment method in 100 experiment times

the time for processing one image was about 70 ms. The parameters in the control system were set as follows: the sample time was set 100 ms, $\lambda = 16$, and the proportional, integral, and differential gains of the PID controller were set as 0.45, 0.12, and 0.04, respectively.

The target and current initial points of the wheel hubs were defined as follows. Firstly, the wheel hub was rotated by hand and the initial point of the wheel hub was chosen carefully. The vision sensor was triggered and an image of the wheel hub was captured. The initial point of the wheel hub was computed, which was served as the target initial point. Since the robot polishing path was the same for the same type of the wheel hubs, the target initial point was computed only once for the first wheel hub and was kept constant for the wheel hubs with the same type. Then, the current initial point of the wheel hub was computed, which was carried out for every wheel hub. Since the current initial point of the wheel hub was used as the feedback signal in the closed-loop controller, it was computed at each control sample time.

The results of the proposed initial point alignment method for the wheel hubs are shown in Figs. 9, 10, and 11. Figure 9 shows the error changes and output of the controller in the initial point alignment control for one wheel hub. The original error was 0.5406 rad. The error converged in about 25 control periods. To evaluate the precision of the proposed initial point alignment further, about 100 experimental times of initial point alignment were conducted. The target initial point was marked. After the initial point alignment was completed, the distance between the polishing tool head and the marked target initial point was measured with a caliper. The error changes of the initial point alignment in the experiment are shown in Fig. 10, which shows that the maximum error was about 0.1 mm. The relative position of the marked initial point and the polishing tool head is shown in Fig. 11 in one experiment. Figure 11a shows the relative position of the marked initial point and the polishing tool head before the initial point was aligned while Fig. 11b shows the one after the initial point was aligned. It can be seen that the initial point was well aligned with the proposed initial point alignment control method.

To better evaluate the performance of the proposed initial point alignment method, its efficiency and precision were

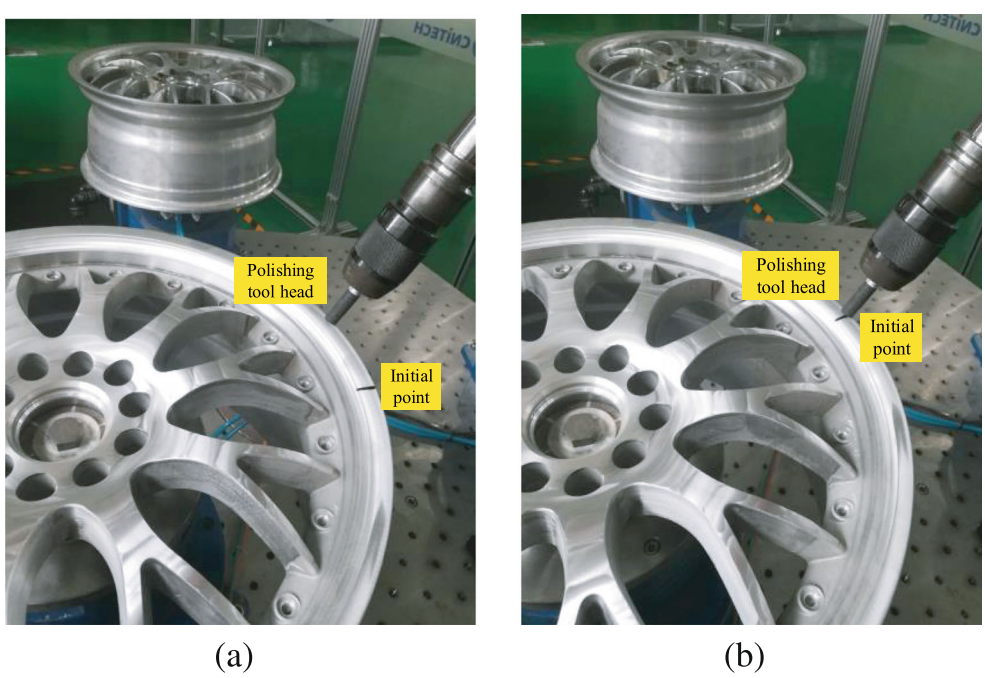


Fig. 11 The results of the proposed initial point alignment control: **a** before the initial point is aligned, **b** after the initial point is aligned

Table 1 Evaluation of the efficiency and precision of the proposed initial point alignment method

Method	Application field	Time cost (ms)	Precision (mm)
Ref. [18]	Welding	52	0.18
Ref. [19]	Welding	94	0.23
This paper	Polishing	70	0.1

compared with those proposed in Refs. [18] and [19]. The results are shown in Table 1. It can be seen that even though the surface shapes of the wheel hubs are much more complex than the seams, the time cost of the proposed method was slightly more than that in Ref. [18], but was less than that in Ref. [19]. The precision of the proposed method was better than that in Refs. [18] and [19]. The comparison experimental results demonstrate the effectiveness of the proposed method further.

7 Conclusion

Initial point alignment is an important process before the robot polishing in the robotic polishing system. In order to align the initial point efficiently and accurately, a non-contact vision sensor is used and a vision-based initial point alignment method is presented in this paper. The difference between the current and target initial points of the wheel hub is denoted as the needed rotation angle of the wheel hub, which can be computed by the center point and the center lines of the spokes of the wheel hub. Therefore, the method to precisely detect the center point and the center lines of the spokes is firstly presented. The center point of the wheel hub is computed based on the GSP method. Then, the center lines of the spokes of the wheel hub are extracted based on the edges of the spokes. From the center point and the center lines of the spokes above, the needed rotation angle of the wheel hub denoting the current and target initial points can be computed. After the needed rotation angles of the wheel hub are computed, a closed-loop initial point alignment control system is designed to align the current initial point of the wheel hub to the target one. Full experiments on several types of wheel hubs with different initial point deviations are well conducted. The results show that the initial point alignment errors can be controlled within 0.1 mm with the proposed method, which can well meet the demand of the robotic polishing system for the wheel hubs.

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References

1. Luo SJ, Fu YT, Zhou YX (2012) Perceptual matching of shape design style between wheel hub and car type. Int J Ind Ergon 42(1):90–102
2. Ehsan F, Mahdi M, Paul DK, Homer R, Gareth T, Alan W, Robert W (2017) Effect of mesh phasing on the transmission efficiency and dynamic performance of wheel hub planetary gear sets. Proceedings of the Institution of Mechanical Engineers, Part C: Journal of Mechanical Engineering Science 232(19):3469–3481
3. Osma A (2009) A statistical approach to comparing wheel hub forging processes. Proceedings of the Institution of Mechanical Engineers, Part D: Journal of Automobile Engineering 223(12):1559–1576
4. Tian FJ, Li ZG, Lv C, Liu GB (2016) Polishing pressure investigations of robot automatic polishing on curved surfaces. Int J Adv Manuf Technol 87(1-4):639–646
5. Wang GL, Wang YQ, Zhang L, Zhao J, Zhou HB (2014) Development and polishing process of a mobile robot finishing large mold surface. Mach Sci Technol 18(4):603–625
6. Ahmed K, Du TT, Zhang YJ (2017) A practical approach for automated polishing system of free-form surface path generation based on industrial arm robot. Int J Adv Manuf Technol 93:3921–3934
7. Mohammad A, Hong J, Wang D (2018) Design of a force-controlled end-effector with low-inertial effect for robotic polishing using macro-mini robot approach. Robot Comput Integr Manuf 49:54–65
8. Ren CX, Gong YQ, Jia F, Wang XS (2016) Theoretical analysis of a six-axis force/torque sensor with overload protection for polishing robot. In: Proceeding of the 23rd international conference on mechatronics and machine vision in practice, 28-30 Nov. 2016, Nanjing, China
9. Lin FY (2014) Path generation for robot polishing system based on cutter location data. Advanced Materials Research 902:250–253
10. Liu J, Huang XL, Fang SW, Chen HP, Xi N (2016) Industrial robot path planning for polishing applications. In: Proceedings of the IEEE international conference on robotics and biomimetics, 3-7 Dec. 2016, Qingdao, China
11. Zhang XD, Chen H, Yang N, Lin H, He K (2017) A structure and control design of constant force polishing end actuator based on polishing robot. In: Proceedings of the IEEE international conference on information and automation, 18-20 July 2017, Macau, China
12. Mohammad A, Hong J, Wang DW, Guan YS (2019) Synergistic integrated design of an electrochemical mechanical polishing end-effector for robotic polishing applications. Robot Comput Integr Manuf 55:65–75
13. Mohsin I, He K, Cai J, Chen H, Du R (2017) Robotic polishing with force controlled end effector and multi-step path planning. In: Proceedings of the IEEE international conference on information and automation, 18-20 July 2017, Macau, China
14. Xie XH, Sun LN (2016) Force control based robotic grinding system and application. In: Proceedings of the 12th world congress on intelligent control and automation, 12-15 June 2016, Guilin, China
15. Feng DY, Sun YW, Du HP (2014) Investigations on the automatic precision polishing of curved surfaces using a five-axis machining centre. Int J Adv Manufact Technol 72:1625C1637

16. Agustina BD, Marin MM, Teti R, Rubio EM (2014) Surface roughness evaluation based on acoustic emission signals in robot assisted polishing. Sensors 14(11):21514–21522
17. Segreto T, Karam S, Teti R (2017) Signal processing and pattern recognition for surface roughness assessment in multiple sensor monitoring of robot-assisted polishing. Int J Adv Manufact Technol 90:1023–1033
18. Fang ZJ, Xu D, Tan M (2013) Vision-based initial weld point positioning using the geometric relationship between two seams. Int J Adv Manufact Technol 66:1535–1543
19. Zhu ZY, Lin T, Chen SB (2005) Recognition of the initial position of weld based on the image pattern match technology for welding robot. Int J Adv Manufact Technol 26:784–788
20. Rafael CG, Richard EW (2017) Digital image processing. Publishing House of Electronics Industry, Beijing
21. Otsu N (1979) A threshold selection method from gray-level histograms. IEEE Trans Sys Man Cybern 9(1):62–66

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